- Shift Scheduling (daily, period < day)
  - Entities
    - Planning horizon
      - T_{\max}
    - Set of periods (with time increment)
      - T = {1,2,...t, .. T_{\max}/t_size} with t_size determined by the modeller
    - Shifts
      - Set of shifts (given)
        - S = {s}
      - Set of shifts (hidden)
    - Breaks
      - Types of breaks (first breaks, 2nd breaks, ..., lunch breaks)
        - G = {1,2,...,L}
      - Sets of first breaks, 2nd breaks, ..., lunch breaks (usually hidden)
        - B = \cup B_g, g \in G
          - B_g is constructed based on requirements
  - Parameters
    - Shifts
      - Start time of each shift
        - start_s \forall s \in S
      - (Regular) Duration of each shift
        - duration_s \forall s \in S
      - Overtime of each shift (optional)
        - ot_s \forall s \in S
    - Breaks
      - Eligibility of breaks
        - Minimum duration eligible for 1 break
          - el_1
        - Minimum duration eligible for 2 breaks
          - el_2
        - ...
        - Minimum duration eligible for lunch break
          - el_L
      - Duration of different breaks for different shifts
        - d_{g,s} \forall g \in G, s \in S (hidden part: d_{g,s} = 0 if duration_s < el_g)
      - Time window of 1st, 2nd, ..., lunch breaks (earliest and latest time a break can start)
        - (e_{g,s}, l_{g,s}) \forall g \in G
          - A series of calculations are required to convert time windows of shift types to shifts
      - Start time of each break (may be hidden)
        - st_{b,g,s} = e_{g,s}, e_{g,s}+1, ..., l_{g,s} \forall g \in G, s \in S, each start time contributes to a break b in B_g
    - Demands
      - Minimun number of employees required per period
        - Shift independent
          - lb_t \forall t \in T
        - Shift dependent
          - lb_{t,s} \forall t \in T, s \in S
      - Exact number of employees required per period
        - Shift independent
          - exact_t \forall t \in T
        - Shift dependent
          - exact_{t,s} \forall t \in T, s \in S
      - Maximum number of employees required per period
        - Shift independent
          - ub_t \forall t \in T
        - Shift dependent
          - ub_{t,s} \forall t \in T, s \in S
    - Costs
      - Cost per shift
        - Fixed for each shift
          - c_s \forall s \in S
        - Duration dependent
          - Break independent
            - c_s = c^R * duration_s + c^O * ot_s \forall s \in S
          - Break dependent (break time not paid)
            - c_s = c^R * duration_s + c^O * ot_s - \sum_{g \in G} c^R * d_{g,s} \forall s \in S
      - Cost per period
        - Regular time
          - c^R
        - Overtime
          - c^O
    - Maximum number of employees on break
      - h_t \forall t \in T
    - Binary parameter indicating if shift s covers period t (1) and otherwise (0) (hidden)
      - a_{s,t} = 1 for start_s \leq t < start_s + duration_s
      - a_{s',t} = 1 for start_s \leq t < start_s + duration_s + over_time_s
    - Binary parameter indicating if break b covers period t (1) and otherwise (0) (hidden)
      - w_{t,b,g,s} = 1 for st_{b,g,s} \leq t < st_{b,g,s} + d_{g,s} if d_{g,s} > 0
  - Decision variables
    - Number of staff for each shift
      - x_s \forall s \in S
    - Number of staff on break (type dependent) for each shift at each period
      - y_{b,g,s} \forall s \in S, b \in B_g, g \in G
  - Objectives
    - Minimisation
      - Minimising total local cost
        - \min\ \sum_{s \in S} c_s x_s
      - Minimising total number of employees =
        - \min\ \sum_{s \in S} x_s
  - Constraints
    - Demand satisfaction
      - Minimum number of employees
        - Shift independent
          - \sum_{s \in S} (a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s}) \geq lb_t \forall t \in T
        - Shift dependent
          - a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s} \geq lb_{t,s} \forall t \in T
      - Exact number of employees
        - Shift dependent
          - a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s} = exact_{t,s} \forall t \in T
        - Shift independent
          - \sum_{s \in S} (a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s}) = exact_t \forall t \in T
      - Maximum number of employees
        - Shift dependent
          - a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s} \leq ub_{t,s} \forall t \in T
        - Shift independent
          - \sum_{s \in S} (a_{s,t} * x_s - \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} * y_{b,g,s}) \leq ub_t \forall t \in T
    - Limit on the employees on break
      - \sum_{s \in S} \sum_{g \in G} \sum_{b \in B_g} w_{t,b,g,s} y_{b,g,s} \leq h_t \forall t \in T
    - Integrality constraints (hidden)
      - x_s \in \{0,1,2,...} \forall s \in S
      - y_{b,g,s} \in \{0,1,2,...} \forall s \in \S, t \in T, g \in G